

MedTrack DV

Hospital Operations & Patient Analytics Dashboard

FINAL PROJECT DOCUMENTATION

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Tableau Desktop / Tableau Public | Source: Fact_Admissions_Clean.csv

Document Control

Item	Details
Project Name	MedTrack DV – Hospital Operations & Patient Analytics Dashboard
Document	Final Project Documentation
Primary Platform	Tableau Desktop / Tableau Public
Data Processing	Python, Pandas, NumPy
Final Analytics Asset	Fact_Admissions_Clean.csv + supporting source datasets
Final Dashboard Asset	MedTrack_DV.twbx
Dashboard Count	4 interconnected dashboards
KPI Count	11 implemented KPIs
QA Result	11 / 11 KPIs validated; 0 calculation mismatches
Dashboard Testing Result	PASS — no critical dashboard issue identified

1. Executive Summary

MedTrack DV is a hospital operations and patient analytics dashboard solution designed to transform healthcare operational data into structured, interactive insights for monitoring hospital performance, patient admissions, department activity and resource utilization. The project uses multiple hospital datasets, applies data cleaning and relationship-aware integration, engineers operational KPIs and presents the results through four interconnected Tableau dashboards.

The final solution follows the project workflow of data collection, data cleaning and transformation, KPI engineering, dashboard planning, dashboard development, dashboard integration, testing and validation, and documentation and delivery.

The completed analytical layer contains 11 implemented KPIs. The separate KPI QA exercise independently reconciled all 11 KPI values, with no calculation mismatches. The dashboard testing exercise reviewed the four-dashboard structure, KPI presentation, visual organization, consistency and functional integration evidence.

2. Project Background

Hospital operations generate large volumes of patient, admission, department, ward, bed and workforce data. Without an integrated analytical view, operational information can remain distributed across multiple datasets, making it difficult to monitor patient flow, capacity, department activity and resource availability consistently.

MedTrack DV addresses this requirement by integrating relevant healthcare datasets, preparing a Tableau-ready analytical layer and providing an interactive dashboard suite for operational monitoring and data-driven decision-making. The project statement specifically identifies hospital performance, patient admissions, department efficiency and healthcare resource utilization as core analytical areas.

3. Problem Statement

Hospital operational data is distributed across multiple entities and contains relationships that can create duplication if datasets are merged without understanding their grain. A reporting solution must therefore establish a reliable analytical structure, maintain correct admission-level metrics, and provide decision-makers with a clear view of hospital activity.

The key challenge addressed by MedTrack DV is to convert these operational datasets into a reliable, interactive and decision-oriented Tableau solution without introducing aggregation errors or losing analytical context.

4. Project Objectives

- Collect and integrate relevant hospital operational and patient admission datasets.

- Understand primary-key, foreign-key and one-to-many relationships before integration.
- Clean and standardize the datasets for analytics.
- Create a unified Tableau-ready admission fact dataset.
- Engineer hospital performance and patient analytics KPIs.
- Develop four interactive Tableau dashboards.
- Provide patient-flow, department and resource-utilization analysis.
- Validate KPI calculations and healthcare metrics.
- Test dashboard interactions and visual usability.
- Deliver a documented, portfolio-ready Tableau dashboard solution.

5. Project Scope

Area	Scope
Data	Hospital operational, patient admission, doctor and supporting healthcare datasets
Data Engineering	Cleaning, standardization, relationship analysis and integration
KPI Engineering	11 implemented operational and patient analytics KPIs
Visualization	Four Tableau dashboards
Interaction	Filters, navigation, dashboard actions and dashboard linking
Testing	KPI QA + dashboard testing
Documentation	Final project documentation + separate QA and dashboard testing reports

6. Data Collection

The project collected 19 hospital datasets in CSV format and imported them into Jupyter Notebook using Pandas. The datasets included Admission, Patient, Department, Ward, Bed, Disease, Doctor, Employee, Billing, Billing Detail, Prescription, DiagnosticTest, Patient Diagnostic, Drug, Drug Inventory, Drug Manufacturer, Patient Insurance, Insurance Provider and Staff Assignment.

The database schema was reviewed before integration to identify primary keys, foreign keys, one-to-one relationships and one-to-many relationships. The Admission table was selected as the central fact table because it connects patients, departments, wards, beds and diseases.

7. Data Cleaning & Transformation

The data preparation stage included structure verification, missing-value checks, duplicate checks, date conversion, data-type standardization and preparation for integration. Pandas and NumPy were used for data processing and cleaning.

- Verified dataset structures and column definitions.
- Checked missing values.
- Checked duplicate records.
- Converted date fields into appropriate datetime formats.
- Standardized data types.
- Prepared datasets for integration and Tableau analysis.

8. Data Integration & Data Model

The one-to-one related datasets—Admission, Patient, Department, Ward, Bed and Disease—were integrated into the unified analytical dataset Fact_Admissions_Clean.csv. This dataset was created as the main Tableau-ready source for hospital analytics.

One-to-many datasets such as Billing Detail, Prescription, Patient Diagnostic, Staff Assignment and Drug Inventory were not directly merged into the admission-level fact because a single admission can have multiple related records. They were instead intended to be connected separately or summarized during KPI engineering to maintain analytical accuracy.

Layer / Entity	Role
Admission	Primary fact / admission-level analytical grain
Patient	Patient attributes and patient-level analysis
Department	Department classification and comparison
Ward	Ward-level operational and capacity context
Bed	Bed availability and utilization context
Disease	Disease/admission analysis
Doctor	Workforce and doctor-count validation
Readmission Summary	Readmission classification and rate validation

9. KPI Engineering

The project specification identifies hospital KPI engineering as a dedicated module. Planned healthcare measures include Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate and Department Efficiency Score.

KPI	Final Value	Primary Validation Logic
Total Patients	23,275	COUNTD(Patient ID)
Total Admissions	45,000	Admission population / COUNT(Admission ID)
Total Departments	6	COUNTD(Department)
Total Wards	27	COUNTD(Ward)
Total Beds	415	Ward capacity aggregated once per distinct ward
Total Doctors	98	COUNTD(Doctor ID)
Available Beds	145	COUNTD(Bed Number)
Occupied Beds	270	Total Beds – Available Beds
Bed Occupancy Rate	65%	Occupied Beds ÷ Total Beds × 100; displayed rounded
Average Length of Stay	5.16 days	AVG(Length of Stay)
Readmission Rate	2.49%	1,120 ÷ 45,000 × 100

10. KPI Validation Outcome

A separate Final QA Testing Report was completed to validate every implemented KPI against the appropriate source data. The final outcome was 11/11 KPIs validated, with zero calculation mismatches.

QA Measure	Result
KPIs tested	11
KPIs passed	11
KPIs failed	0
Calculation mismatches	0
Final KPI QA status	PASS — 11/11 validated

The Readmission Rate was independently validated from the readmission summary: 1,120 records marked Yes and 43,880 marked No, producing a total of 45,000 records and a readmission rate of 2.49%.

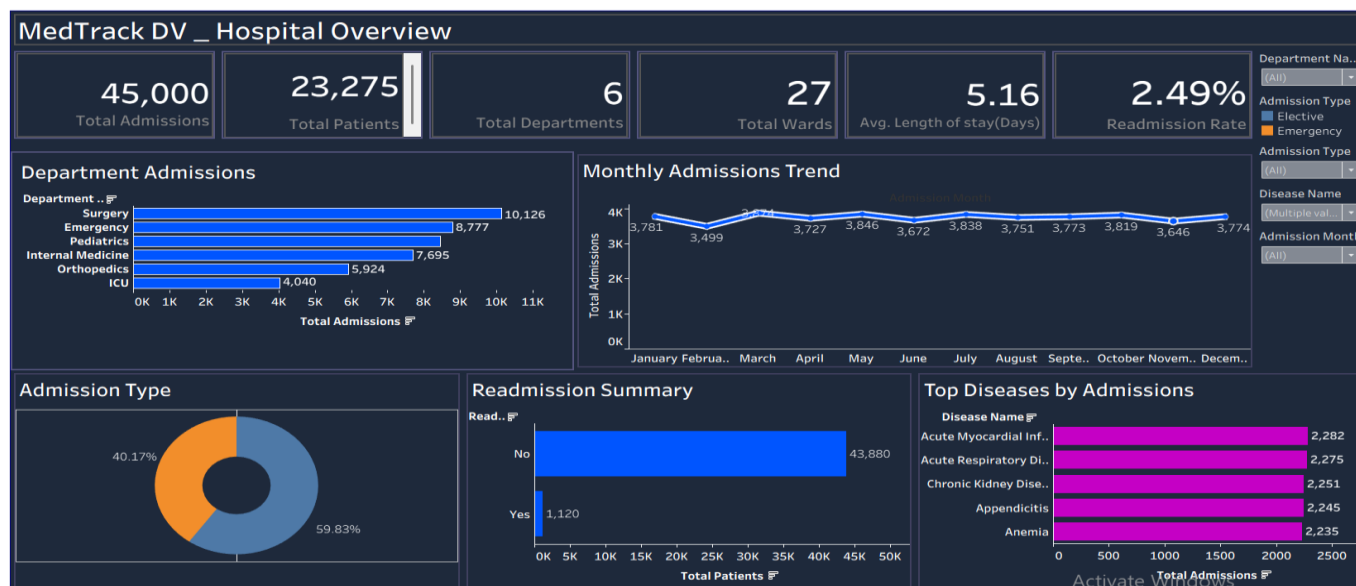
11. Dashboard Architecture

The final Tableau workbook is designed as a unified dashboard suite containing four interconnected dashboards: Hospital Overview, Patient Flow, Department Analytics and Resource Utilization. This four-dashboard structure is explicitly defined in the project specification.

Dashboard	Purpose	Key Analytical Areas
Hospital Overview	Executive operational summary	Admissions, occupancy, readmission, hospital KPIs, monthly trends
Patient Flow	Patient movement and admission analysis	Admission trends, discharge tracking, patient movement, LOS, peak load
Department Analytics	Department performance comparison	Patient volume, readmission, efficiency, capacity, LOS and disease analysis
Resource Utilization	Hospital capacity and workforce analysis	Beds, bed status, doctor specialization, staffing/resource availability

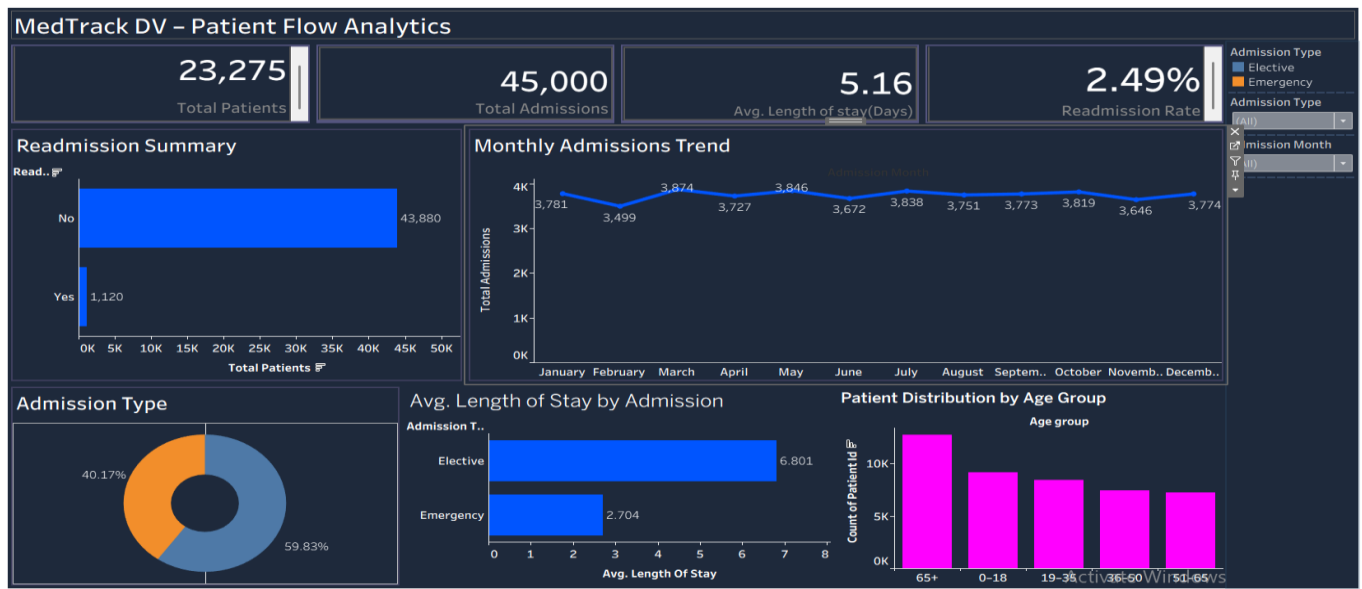
12. Hospital Overview Dashboard

The Hospital Overview dashboard provides the high-level operational view. It is designed to present hospital KPIs and supporting analytical visuals for admissions, occupancy, readmission and operational trends. The project scope specifically identifies admissions overview, occupancy monitoring, readmission analysis, hospital performance KPIs and monthly operational trends for this dashboard.



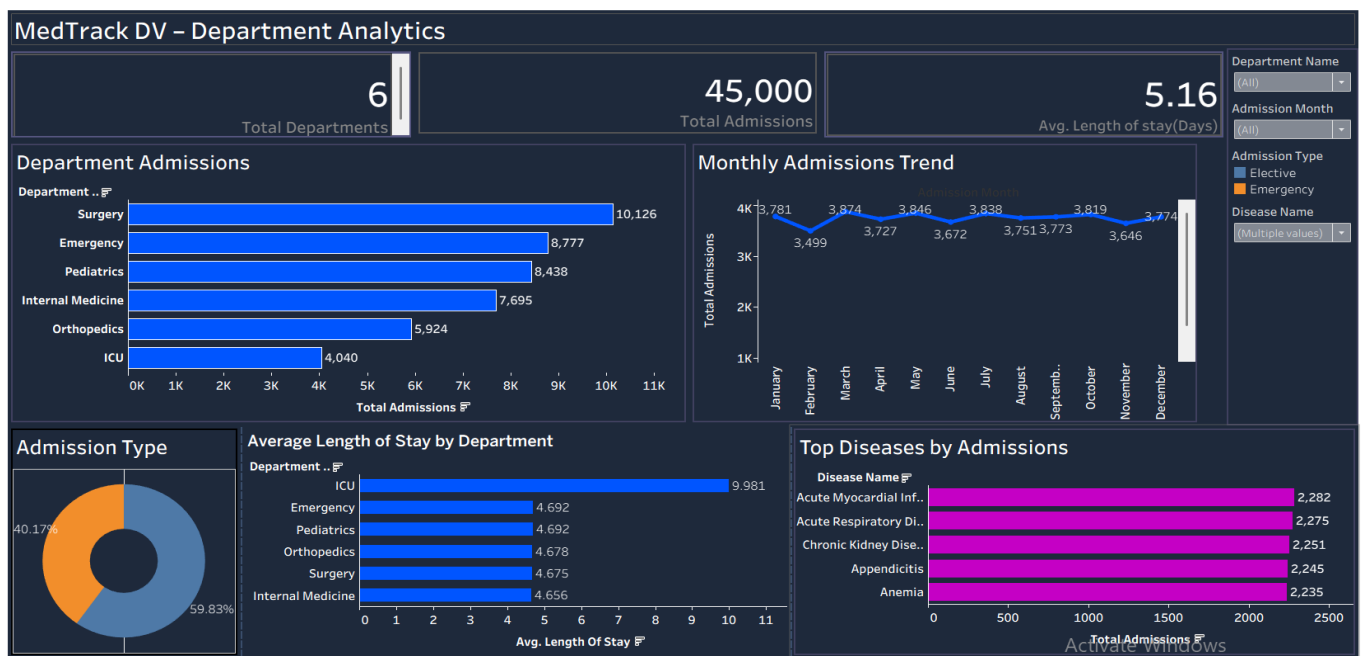
13. Patient Flow Dashboard

The Patient Flow dashboard focuses on how patients enter, move through and are represented within the hospital admission process. Its intended analysis includes admission trends, discharge tracking, patient movement, average stay and peak patient load monitoring.



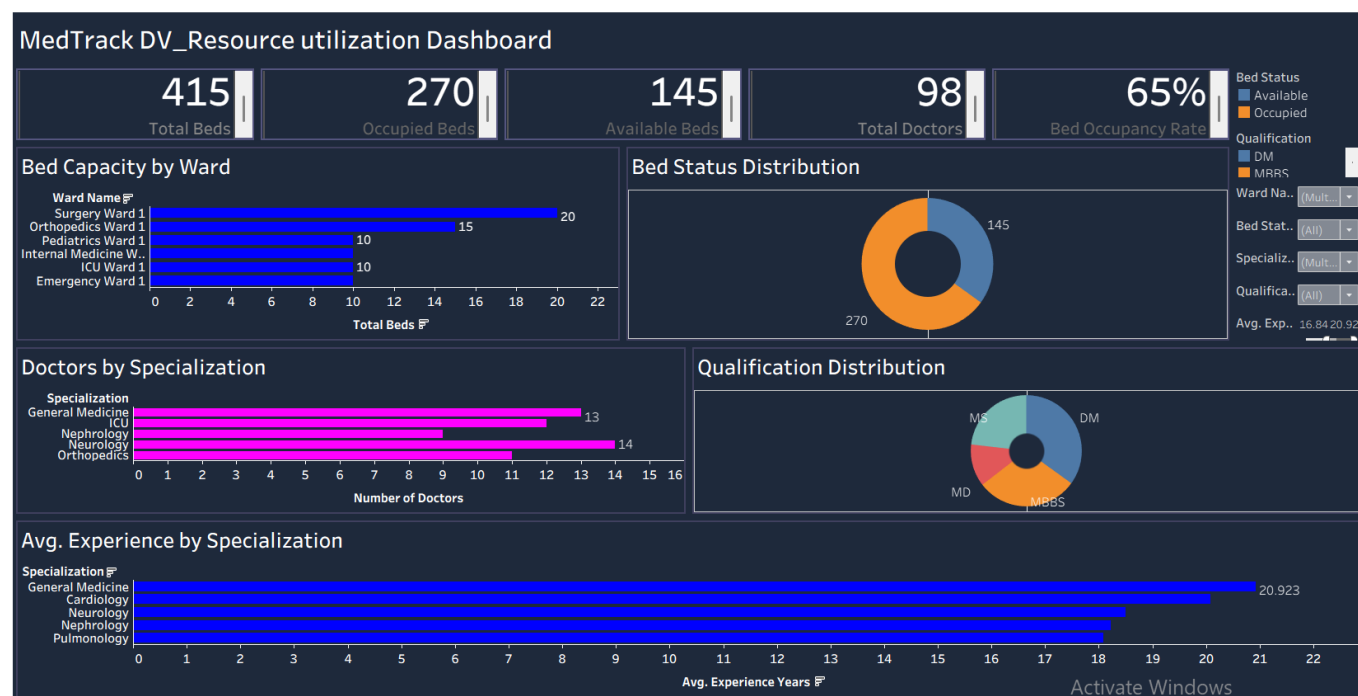
14. Department Analytics Dashboard

Department Analytics provides comparative analysis of hospital departments. The project scope includes department performance, patient volume by department, readmission by department, department efficiency comparison and treatment capacity analysis.



15. Resource Utilization Dashboard

Resource Utilization focuses on hospital capacity and workforce characteristics. The planned scope includes bed utilization, staff allocation, equipment utilization, and capacity planning and resource availability analysis.



16. Dashboard Integration

The dashboard suite is designed as an integrated Tableau experience. Global filters, navigation controls, parameter actions and dashboard linking are part of the specified integration layer. The project evidence states that the dashboards are integrated and that navigation and filters are functioning correctly.

17. Technology Stack

Area	Technology / Tool
Data Collection	Python, hospital datasets, patient admission datasets
Data Processing	Pandas, NumPy
Data Cleaning	Python
Visualization	Tableau Desktop / Tableau Public
Dashboard Integration	Tableau Filters, Parameters, Actions
Documentation	Markdown, GitHub

18. Testing & Validation Strategy

Testing was separated into two formal deliverables: KPI QA Testing and Dashboard Testing. This separation ensures that numerical correctness and dashboard usability/functionality are assessed independently.

Testing Layer	Purpose	Outcome
KPI QA Testing	Source-to-dashboard calculation reconciliation	PASS — 11/11 KPIs validated
Dashboard Testing	Structure, KPI presentation, visuals, usability and integration	PASS — no critical issue identified

19. Dashboard Testing Outcome

The separate Dashboard Testing Report reviewed the four dashboards for structure, KPI presentation, visual quality, layout, readability, cross-dashboard consistency and functional integration evidence. No critical or high-severity dashboard issue was identified.

Assessment	Result
Dashboards reviewed	4
KPI presentation reviewed	11 KPIs
Critical defects	0
High-severity defects	0
Final dashboard test status	PASS

20. Key Analytical Capabilities

- Hospital-level KPI monitoring.
- Admission-volume and monthly trend analysis.
- Patient-flow and admission-type analysis.
- Readmission monitoring.
- Average length-of-stay analysis.
- Department-level admission and performance comparison.
- Ward and bed-capacity monitoring.
- Bed availability and occupancy analysis.
- Doctor-count and workforce analysis.
- Disease/admission pattern analysis.

21. Data Quality & Governance Considerations

- Maintain admission-level grain for admission KPIs.
- Use distinct counts for business entities such as patients, departments, wards, beds and doctors.
- Avoid directly joining one-to-many datasets to the admission fact without aggregation.
- Aggregate ward capacity once per ward to avoid capacity inflation.
- Maintain separate supporting datasets where the business grain differs.
- Validate KPI calculations whenever source data or business rules change.
- Retain QA evidence alongside the final Tableau workbook.

22. Project Deliverables

Deliverable	Purpose / Status
Fact_Admissions_Clean.csv	Unified Tableau-ready analytical dataset
MedTrack_DV.twbx	Final Tableau dashboard workbook
Final QA Testing Report	11-KPI source reconciliation
Dashboard Testing Report	Dashboard functional/visual validation
Final Project Documentation	End-to-end project report
GitHub Repository	Planned project delivery/portfolio structure
Tableau Public	Optional deployment identified in project specification

The project specification identifies the final documentation, Tableau workbook and GitHub repository as delivery artifacts, with Tableau Public listed as optional.

23. Project Structure

Folder / Asset	Recommended Content
/data	Raw and cleaned CSV datasets
/scripts	Data collection, cleaning and KPI scripts
/dashboard	MedTrack_DV.twbx and dashboard-related assets

The /scripts, /data, /dashboard and /docs structure is explicitly identified in the project delivery plan.

24. Limitations

This documentation describes the implemented solution and the testing outcomes supported by the supplied project materials. It does not claim live production deployment, real-time hospital-system integration, clinical decision support, or patient-level clinical recommendations.

The dashboard is an analytics and operational monitoring solution based on the supplied datasets. Business users should interpret results within the data definitions and time coverage of the source datasets.

25. Future Enhancements

- Add automated data-refresh pipelines for recurring dashboard updates.
- Introduce role-based dashboard views for hospital management, department managers and resource teams.
- Expand operational alerts for capacity and occupancy thresholds.
- Add deeper department efficiency and treatment-capacity measures when validated source fields are available.
- Add more granular time-based patient-flow analysis.
- Extend resource analytics to equipment utilization when reliable source data is available.
- Automate KPI QA checks as part of the data-refresh process.

26. Final Project Outcome

MedTrack DV successfully establishes a structured healthcare analytics workflow from multi-dataset collection and cleaning through relationship-aware data integration, KPI engineering, Tableau visualization, testing and final documentation. The solution delivers four interconnected dashboards covering hospital overview, patient flow, department analytics and resource utilization.

The final KPI QA confirms that all 11 implemented KPIs are validated with zero calculation mismatches. The separate dashboard testing exercise also identified no critical dashboard issues. The project therefore satisfies the defined analytical, dashboard and validation objectives and is ready for final submission, subject to the user's final Tableau workbook and evidence package being the submitted versions.

FINAL PROJECT STATUS: COMPLETED — READY FOR FINAL SUBMISSION

27. References & Project Traceability

This report is grounded in the supplied MedTrack DV project specification and progress materials. The project specification defines the dashboard scope, workflow, KPI engineering, integration requirements, testing requirements and documentation deliverables.

The project progress presentation documents the collection of 19 datasets, relationship analysis, creation of Fact_Admissions_Clean.csv and the treatment of one-to-many datasets.

28. Final Sign-Off

Sign-Off Area	Status
Data collection & preparation	Completed
Data cleaning & transformation	Completed

Data integration	Completed
KPI engineering	Completed
Dashboard development	Completed
Dashboard integration	Completed
KPI QA testing	PASS — 11/11
Dashboard testing	PASS
Final documentation	Completed
Final recommendation	APPROVED FOR FINAL SUBMISSION